

Executive Summary

PULSAR AI is an AI-powered pediatric pneumonia detection and clinical management platform developed at the University of Molay Taher Saïda, Algeria. The system integrates a Gradient Boosting classifier for vital signs analysis, a DenseNet121 convolutional neural network for chest X-ray interpretation, and a Retrieval-Augmented Generation (RAG) chatbot for clinical decision support — forming a complete clinical workflow system covering diagnosis, treatment planning, and automated medical documentation.

This report presents a significantly enhanced business model built on rigorous technical and financial analysis specific to the Algerian healthcare ecosystem. It covers: a six-card value proposition balancing medical and commercial dimensions, detailed cost analysis across three deployment scales, competitive landscape assessment, a phased go-to-market strategy, and a restructured business model following the KHIAT Startup framework — including revenue streams, pricing in DZD, customer segments, and scalability logic aligned with Algeria's healthcare digitalization trajectory.

1. Value Proposition

The value proposition is organized into two tiers: three cards capturing the clinical and medical value delivered to physicians and patients, and three cards capturing the business and institutional value created for healthcare organizations and stakeholders.

MEDICAL VALUE

Dual-Pathway Diagnostic Precision

GBM classifier on 12 vital-sign parameters (confusion: 64.8% feature importance) combined with DenseNet121 CNN (95% sensitivity) and GradCAM visual explainability — providing evidence-grounded pneumonia detection for children aged 1–16.

RAG-Powered Clinical Decision Support

A hybrid FAISS + BM25 retrieval engine queries both patient records and a curated medical knowledge base in real time. Structured prompts enforce clinical reasoning rules, delivering risk identification, treatment failure detection, and evidence-cited recommendations in under 200 words per query.

Automated Medical Documentation

Full patient records, treatment plans, monitoring tables, and AI-generated reports are produced automatically — eliminating transcription overhead, ensuring audit-trail compliance, and enabling seamless handover without manual documentation.

BUSINESS VALUE

Predictable Cost Efficiency

Storage costs are negligible (iBox 100 GB / 8,000 DZD/year covers all scales). AI inference is the sole variable cost — controlled via query caching and single-call report generation, enabling 3× to 5× cost reduction while maintaining clinical quality.

Scalable SaaS Architecture

A single codebase serves cabinets (30 pts/day), clinics (100 pts/day), and hospitals (200 pts/day) with marginal incremental infrastructure cost. The file-based persistence model deploys on any machine with no database server — critical for resource-constrained Algerian clinic environments.

Competitive Differentiation through Data

The structured medical data lake (Chest X-rays, diagnostics, treatment PDFs) constitutes a monetizable asset — serving research labs and universities from Year 1, powering a specialized Algerian medical research platform from Year 3 — creating compounding competitive barriers no international player can replicate locally.

2. Market Analysis — Algeria (Unchanged)

48M

1,200+

Population Algeria

5,000+

Public Hospitals

15+

Private Clinics

Medical Universities

Market Segment	2024 Value (DZD)	CAGR (2024–2030)	Context
Healthcare IT & EMR Systems	~40 Milliard DZD	~12% / year	Hospital digitalization ongoing (Plan Santé 2025)
Clinical Decision Support (CDSS)	~12 Milliard DZD	~15% / year	Growing adoption in CHUs and private clinics
Pediatric & Emergency Market	~18 Milliard DZD	~10% / year	Network of 1,200+ hospitals, 5,000+ private clinics
Medical Research & Data (universities, labs)	~5 Milliard DZD	~18% / year	Rising demand for real clinical datasets for research

Algeria's healthcare digitalization is accelerating under the Plan Santé 2025, with the MSPRH mandating electronic health records in public hospitals. The pediatric segment represents a structurally underserved niche: pneumonia remains among the leading causes of pediatric mortality in MENA, and access to AI-assisted radiology is unavailable in most Algerian public facilities outside major CHUs.

3. Cost Analysis in Algeria

The following cost model is derived from the system's technical architecture and real Algerian cloud infrastructure pricing. Two cost categories govern total operational expenditure: cloud storage (negligible) and AI inference via the RAG agent (dominant). All DZD figures are based on market rates as of 2025.

3.1 System Infrastructure Components

Component	Description	Volume	Cost Estimate
Vector Database (FAISS)	Patient PDF knowledge base + Medical KB	~2–3 GB	~10–50 DZD/month
Medical Knowledge Base Embedding	2 GB PDF corpus (~300–600M tokens)	One-time	800–1,600 DZD (one-time)
Structured Patient Data (CSV)	10–50 KB per patient	Negligible	Covered by iBox
Medical PDFs & X-ray Images	~5 MB per patient (avg, including 1 X-ray)	Scales with patients	Covered by iBox
iBox — Algérie Télécom Cloud	100 GB sovereign hosting	Annual flat fee	8,000 DZD/year
AI Inference (OpenRouter / LLM)	4 queries/patient, ~6K–9K tokens/query	Variable	Dominant cost driver

3.2 Monthly Storage Requirements by Scale

Scale	Daily Volume	Monthly Volume	Storage Required	Monthly Storage Cost (DZD)
Cabinet	30 patients/day	900 patients/month	4.5 GB/month	~15–60 DZD
Clinic	100 patients/day	3,000 patients/month	15 GB/month	~50–200 DZD
Hospital	200 patients/day	6,000 patients/month	30 GB/month	~100–400 DZD

Strategic decision: the iBox 100 GB annual subscription (8,000 DZD/year) covers all three scales simultaneously. Storage cost is effectively zero on a per-patient and per-month basis.

3.3 AI Agent (RAG System) Monthly Cost by Scale

Scale	Monthly Patients	Token Volume	USD Estimate	DZD Estimate
Cabinet	900 patients	~27M tokens/month	\$15–\$60	2,000–8,000 DZD
Clinic	3,000 patients	~90M tokens/month	\$50–\$180	7,000–25,000 DZD
Hospital	6,000 patients	~180M tokens/month	\$100–\$400	14,000–55,000 DZD

Token calculation basis: 4 queries/patient × ~6,000–9,000 tokens/query (including patient context ~1K tokens, retrieved RAG context ~2K–5K tokens, prompt ~1K tokens, response ~1K–2K tokens). Embedding the 2 GB medical PDF corpus is a one-time cost of 800–1,600 DZD.

3.4 Total Monthly Operational Cost Summary

Scale	Storage Cost	AI Inference Cost	Total Monthly OpEx	Subscription Price
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Cabinet	~0 DZD (iBox covered)	2,000–8,000 DZD	2,000–8,000 DZD	13,000 DZD
Clinic	~0 DZD (iBox covered)	7,000–25,000 DZD	7,000–25,000 DZD	32,000 DZD
Hospital	~0 DZD (iBox covered)	14,000–55,000 DZD	14,000–55,000 DZD	64,000 DZD

3.5 Optimization Strategies

Response Caching: Identical or near-identical queries (common clinical patterns) are cached, avoiding redundant LLM calls. Estimated 40–60% cache hit rate in clinical practice.

Single-Call Report Generation: The PDF report generator consolidates all patient data into one comprehensive LLM call instead of 4 individual queries — reducing per-report AI cost by ~75%.

Tiered Model Selection: Lightweight models (e.g., Mistral-7B via OpenRouter) handle routine queries; large models (GPT-4, Claude) are reserved for final diagnostic report generation — reducing average token cost by 2×–3×.

Query Reduction: Reducing default queries from 4 to 2 per patient approximately halves AI costs with minimal clinical quality impact for straightforward cases.

Combined, these optimizations can reduce total AI operational cost by **3× to 5×**, improving unit economics significantly at scale and enabling margin expansion as the subscriber base grows.

4. Competitive Landscape

PULSAR AI operates in a nascent but rapidly growing Algerian health-tech market. The competitive landscape is characterized by the near-absence of locally developed AI clinical decision support tools, the dominance of imported generic HIS platforms, and a structural data advantage for any locally grounded solution.

Criterion	PULSAR AI	International EMR Platforms (e.g., Cerner, OpenMRS)	Generic Telemedicine Apps	Traditional Clinical Practice
AI Diagnostic Engine	■ Dual AI (GBM + DenseNet121)	■ None	■ Basic triage only	■ None
RAG Clinical Chatbot	■ Hybrid FAISS+BM25 + reranker	■ None	■ None	■ None
Algerian Context	■ Built for Algeria (DZD, DZ law)	■ Foreign pricing/compliance	■ Partial	■ Yes
X-Ray GradCAM	■ Visual explainability	■ None	■ None	■ None
PDF Report Generation	■ Automated, timestamped	■ Manual/generic	■ None	■ None
Data Sovereignty	■ iBox (Algérie Télécom)	■ Foreign cloud	■ Foreign cloud	■ Local
Offline/Low-IT Deployment	■ File-based, zero DB	■ Requires server infra	■ Cloud-only	■ Yes
Medical Data Lake (Revenue)	■ Monetizable from Year 2	■ Vendor-locked data	■ None	■ None
Pricing Model (DZD)	■ 13K–64K DZD/month	■ USD pricing (prohibitive)	■ Variable	N/A
LLM Medical Roadmap	■ Year 3 Algerian Medical LLM	■ None locally	■ None	■ None

Key Competitive Differentiation Insights:

Local-first advantage: No international competitor offers DZD-priced, iBox-hosted, Arabized/French-language clinical AI. PULSAR AI removes every adoption barrier — cost, compliance, language, and infrastructure.

Data moat: The structured medical data lake accumulated from Year 1 pilot deployments becomes a defensible competitive asset. Real Algerian pediatric clinical data is unavailable to any foreign competitor.

RAG + AI fusion: The combination of a trained CNN, a gradient boosting classifier, and a RAG chatbot in a single unified clinical workflow has no equivalent in the Algerian market as of 2025.

Regulatory alignment: File-based, locally hosted architecture aligns with emerging Algerian health data protection legislation — a compliance barrier that disadvantages cloud-only foreign platforms.

5. Go-To-Market Strategy

The go-to-market strategy follows a phased rollout model that prioritizes data accumulation and institutional trust before full monetization. This is the optimal approach for a health-tech startup in Algeria where adoption requires demonstrated clinical value and regulatory comfort.

YEAR 1 — FREE PLATFORM Data Collection & Validation

- Deploy platform FREE to hospitals, CHUs, and clinics
- Objective: collect real anonymized clinical data at scale
- Observe system behavior in live clinical conditions
- Establish pilot partnerships with CHU Saïda and regional hospitals
- Build institutional trust and gather clinical feedback
- Develop usage-based pricing from real consumption data
- Academic validation: publish results with University of Saïda

YEAR 2 — DATA MONETIZATION Revenue from Research Institutions

- Platform remains FREE for hospitals and physicians
- Monetize anonymized datasets to research labs and universities
- Clients: USTHB, UDES Oran, pharmaceutical industry (SAIDAL)
- Data licenses: 500,000–5,000,000 DZD per contract
- Expand pilot network to 10+ hospitals across Algeria
- Reinvest revenue into platform improvements and security
- Prepare subscription pricing based on Year 1 usage analytics

YEAR 3+ — FULL MONETIZATION Subscriptions + Medical LLM Launch

- Activate paid subscriptions: physicians, hospitals, pharma
- Launch specialized Algerian Medical LLM as premium service
- Extended data sales to pharmaceutical industry for drug trials
- Deploy on-premise solutions for CHUs and large hospitals
- LLM sold as separate subscription: 8,000 DZD/physician/month
- Target: become the medical OS of Algeria
- Revenue projection: 50–150M DZD (Year 3), 300M+ DZD (Year 4+)

Adoption Strategy Details:

Pilot Program Design: Initial pilots target university hospital affiliates (CHU Saïda) where the academic relationship with the University of Molay Taher Saïda provides institutional access. Each pilot runs 3–6 months with a dedicated clinical coordinator, generating both usage data and peer-reviewed publication potential.

Physician Onboarding: The zero-infrastructure deployment model (no database server required) dramatically reduces IT resistance. Physicians can be operational within hours of profile setup. The intuitive Streamlit interface requires no specialized training beyond a 2-hour onboarding session.

Regulatory Pathway: Engagement with the MSPRH from Year 1 positions PULSAR AI for inclusion in the national Plan Santé 2025 digitalization framework, potentially unlocking public procurement channels and subsidized deployment across public health facilities.

6. Business Model (KHIAT Startup Framework)

The following business model section is structured according to the KHIAT Startup presentation framework as defined by Dr. Abderrahmane KHIAT, covering subscription tiers, pricing in DZD, customer segments, value capture logic, scalability, and revenue investment allocation.

6.1 Subscription-Based Revenue Model (Active from Year 3)

Because of the critical importance of clinical trust and regulatory comfort in the Algerian healthcare market, the platform operates on a free-first model for Years 1–2. Year 3 activates a tiered SaaS subscription model priced entirely in DZD, eliminating foreign currency risk and aligning with Algerian institutional budget cycles.

Offering	Target Segment	Pricing (DZD)	Included Features
MÉDECIN ESSENTIEL	Individual physician / small cabinet	12,000 DZD/month 120,000 DZD/year	Monitoring up to 50 patients, personal performance ranking, PDF report generation, RAG chatbot access
HÔPITAL PRO ■	Public hospital / mid-size clinic (3–15 physicians)	45,000 DZD/month 480,000 DZD/year	Unlimited patient monitoring, AI emergency protocols, institutional ranking, CSV/PDF export, nurse monitoring tables
CHU / GRAND HÔPITAL	CHU, large institution (15+ physicians)	≥120,000 DZD/month Annual contract	On-premise deployment, HIS API integration, premium medical LLM, dedicated training, SLA guarantee
DONNÉES RECHERCHE	Research labs, universities, pharmaceutical companies	500,000–5,000,000 DZD per contract	Anonymized datasets (CSV, X-rays, reports), API access, exclusive license options, thesis-grade dataset packages
LLM MÉDICAL (Year 3+)	Physicians, hospitals, healthcare industry	8,000 DZD/physician/month Separate subscription	Specialized Algerian medical LLM: diagnostic reasoning, protocols, clinical decision support in Arabic/French/Darija

* Years 1 & 2: Platform is FREE. Revenue derives exclusively from data licensing contracts starting Year 2.

6.2 Customer Segments

Segment	Revenue Start	Key Value Drivers
■■■■■ Physicians & Specialists	Year 3	Performance ranking dashboard, patient monitoring, medical discovery insights, medical LLM subscription
■ Hospitals & Clinics	Year 3	Institutional performance evaluation, resource management, AI emergency protocols, reputation building
■ Research Labs & Universities	Year 2	Anonymized clinical datasets (CSV, X-rays, reports), API access, thesis datasets, disease model correlations
■ Pharmaceutical Industry	Year 3	Drug efficacy analysis, side-effect detection, clinical trial data, exclusive specialized dataset licenses

6.3 Value Capture Logic

PULSAR AI captures value through a two-layer model. The **primary layer** is recurring SaaS subscriptions from healthcare institutions — predictable, high-retention revenue with low churn due to clinical workflow integration. The **secondary layer** is data licensing — a high-margin, near-zero-marginal-cost revenue stream where the asset (anonymized clinical data) appreciates in value as the dataset grows. The two layers are reinforcing: the

free platform maximizes data accumulation, which maximizes data licensing revenue, which funds the platform improvements that maximize subscription conversion.

Phase	Revenue Stream	Projected Revenue	Revenue Drivers
Year 1	Free	0 DZD	Free platform, data collection, pilot partnerships
Year 2	Data Licensing	2–10 Million DZD	Anonymized dataset sales to research labs and universities
Year 3	Subscriptions + Data	50–150 Million DZD	Full subscription activation + expanded data licensing
Year 4+	Subscriptions + Data + LLM	300+ Million DZD	Medical LLM launch, pharma partnerships, CHU on-premise contracts

6.4 Revenue Investment Allocation (Year 3+)

Generated revenues are fully reinvested to build the Algerian medical AI ecosystem:

% Allocation	Investment Area	Purpose
40%	Specialized AI Engineers	Recruit NLP and deep learning engineers for Algerian Medical LLM development
25%	Algerian Medical LLM	Train a large language model specialized for physicians, adapted to Arabic, French, and medical Darija
20%	Infrastructure & Security	Sovereign servers, DZ data protection compliance, secure data lake storage, CHU on-premise deployment
15%	R&D; & Partnerships	University medical collaborations, research labs, MESRS for clinical model validation

6.5 Scalability Considerations

The architecture scales along three independent dimensions:

- Technical scalability:** The file-based, zero-database persistence model deploys on any machine with Python. Horizontal scaling is achieved by simply provisioning additional iBox storage (cost: linear, low). The FAISS vector index scales to millions of chunks without infrastructure changes.
- Economic scalability:** AI inference cost per patient decreases as caching layers mature and model selection improves. The marginal cost of serving the 1,000th subscriber is a fraction of the 10th — creating strong operating leverage as the subscriber base grows.
- Data scalability:** Each new patient record increases the value of the data lake asset. The medical knowledge base expands by adding PDFs to the references directory — no schema changes, no migrations. This creates a compounding data flywheel: more patients → richer dataset → higher data licensing revenue → better models → higher subscription value.
- Geographic scalability:** The DZD pricing model is Algeria-specific, but the architecture is language-agnostic and country-portable. MENA expansion is viable from Year 4 with localized pricing and medical knowledge bases for each target market.

7. Partnerships (Unchanged)



CHU & Public Hospitals

Pilot partnerships from Year 1. Free platform access in exchange for real anonymized clinical data. Clinical validation of the system under real conditions. Pathway to MSPRH endorsement.



Medical Universities (USTHB, UDES Oran, Saïda)

R&D; collaboration, PhD theses, scientific publications based on platform data. Academic credibility that accelerates institutional adoption and provides regulatory support.



Algerian Pharmaceutical Industry (SAIDAL & Private Players)

Access to treatment data for drug research. Revenue from Year 2 via dataset licensing. Potential co-development of disease-specific clinical modules.



CNAS & Health Insurance (Chifa Card Integration)

Integration with the Chifa card reimbursement system. Reduction of costs through better prevention and early detection. Potential to position PULSAR AI as an insurance-recognized clinical tool.



Ministry of Health (MSPRH)

Institutional support. Data for national public health statistics and epidemiological reports. Alignment with Plan Santé 2025. Gateway to public hospital procurement at scale.



Algérie Télécom / iBox

Sovereign, secure cloud infrastructure. Compliance with Algerian health data protection law. Predictable infrastructure cost (8,000 DZD/year for 100 GB). No foreign cloud dependency.

Conclusion & Strategic Summary

PULSAR AI represents a technically rigorous, financially grounded, and strategically sound healthcare AI platform purpose-built for the Algerian context. Its business model combines three defensible value engines: a dual-AI clinical decision support system with documented diagnostic accuracy, a structured medical data lake with compounding monetization potential, and a phased commercialization strategy that builds institutional trust before extracting economic value.

The cost analysis confirms a structurally efficient operational model: storage infrastructure is effectively zero cost, AI inference scales predictably with patient volume, and the margin between operational cost and subscription pricing is sufficient to fund a 4-person technical team expansion within 18 months. The competitive analysis demonstrates a clear first-mover advantage in a market that no international or domestic competitor currently addresses with comparable depth.

The three-year roadmap — free platform for data collection, data licensing for early revenue, full monetization plus Medical LLM launch — mirrors the validated playbook of successful health-tech startups in comparable emerging markets, adapted to the specific regulatory, linguistic, and infrastructural realities of Algeria. With the institutional backing of the University of Molay Taher Saïda and Dr. Abderrahmane KHIAT's supervision, PULSAR AI is positioned to become the foundational medical AI platform of Algeria's healthcare digitalization

decade.

Strategic Dimension	Current Position	Year 3 Target
Deployment Scale	University pilot (CHU Saïda)	50+ hospitals across Algeria
Revenue Model	Free (Year 1) / Data licensing (Year 2)	SaaS + Data + Medical LLM
Revenue Projection	0–10M DZD (Years 1–2)	50–150M DZD (Year 3), 300M+ (Year 4+)
Team Size	4 co-founders (current)	+5 engineers (2 fullstack, 1 security, 1 cloud, 1 data scientist)
Competitive Moat	Dual AI + RAG + local deployment	Data lake + Algerian Medical LLM + institutional contracts